

Project Proposal: Text Detective - Crime Solver

Group Members:

S.No	Name	Roll Number
1	Manaal Amir	24P-0512
2	Zarish Imran	24P-0683
3	Maimoona Naeem	24P-0617

Project Description:

Text Detective: Crime Solver is a terminal-based interactive game where players take on the role of a detective trying to solve a criminal case. The player can interrogate suspects, examine evidence, use hints (at a score cost), and make accusations to solve the case.

Objective:

To design and develop a modular, interactive C++ game using Object-Oriented Programming principles, such as encapsulation, inheritance, polymorphism, and abstraction. The game will demonstrate class structure, code reusability, and a user-friendly text interface.

OOP Concepts Applied:

Concept	Application in Project
Class & Object	Classes like Person, Suspect, Evidence, and Game model real-world entities.
Encapsulation	Data members (e.g., name, age, motive) are kept private and accessed via public methods.
Inheritance	Suspect class inherits from Person class.
Abstraction	Game logic (like case loading, interrogation, hints) is hidden inside member functions.

Polymorphism

Can be extended by adding more cases or types of characters using virtual functions if needed.

Class Breakdown:

- Person: Base class containing name, age, and alibi.
- Suspect: Inherits from Person. Adds motive and guilt status.
- Evidence: Stores description, location, and associated suspect.
- Game: Handles core gameplay – menus, cases, hint logic, score handling, and accusation system.

Features:

- Two unique cases with different suspects and evidence.
- Scoring and attempt tracking.
- Hint system to aid players at a score cost.
- Guilt checking and multiple game outcomes.

Tools & Technologies:

- Language: C++
- Compiler: g++, Code::Blocks, or Visual Studio
- Platform: Console/Terminal

Timeline:

Week	Task
1	Project Planning, Class Design
2	Core Class Implementation (Person, Suspect, Evidence)
3	Game Logic Implementation (Game class)
4	Testing, Debugging, and Enhancements
5	Report Preparation and Final Submission

Expected Output:

- Fully functional terminal-based detective game.
- Modular and expandable C++ code using OOP principles.
- Case-based storylines with multiple possible outcomes.